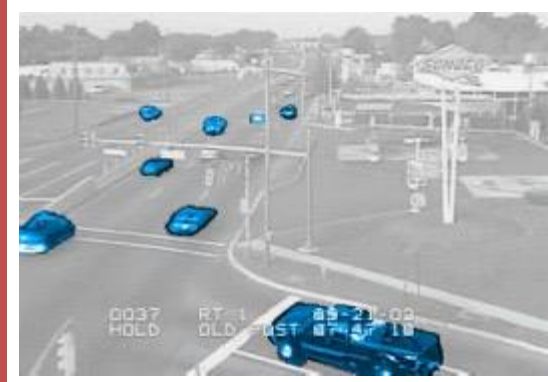


Network Verification

Building a Traffic Simulation

Three ingredients needed:

- **Network** (vehicle, bus, train,...) where movements take place
- **Infrastructure** (traffic lights, toll stations, bus stops,...) and rules to govern traffic behavior
- **Demand** for transport

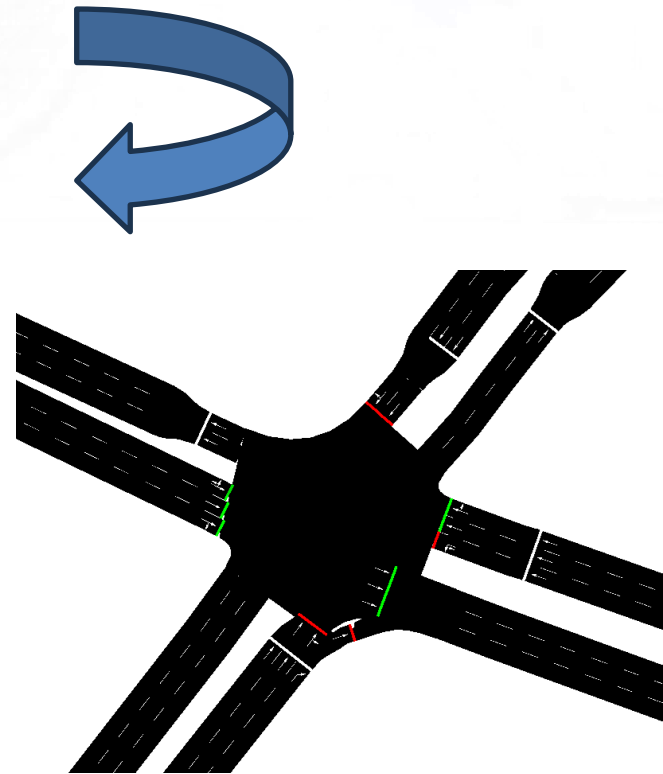
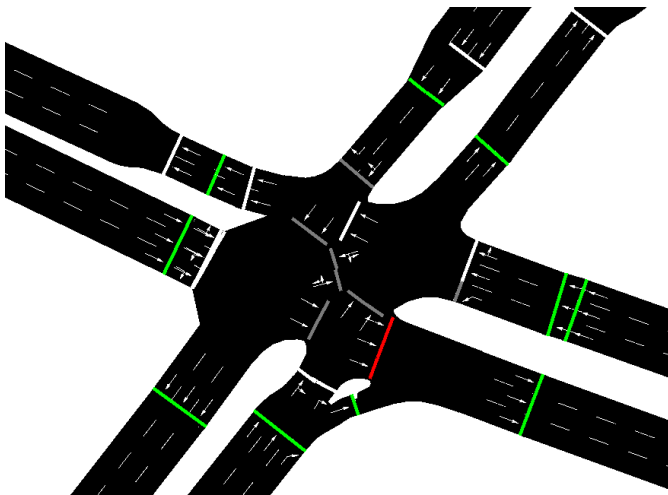


Network Verification Topics

- **Topology:** Roads (edges) and Junctions (nodes)
- **Attributes:** lane count, speed limit
- **Connectivity:** lane-to-lane connections
- **Access:** permissions (allow/disallow)
- **Intersections:** right-of-way
- **Infrastructure:** public transport stops, detectors
- **Traffic lights:** phase layout, timing, actuation

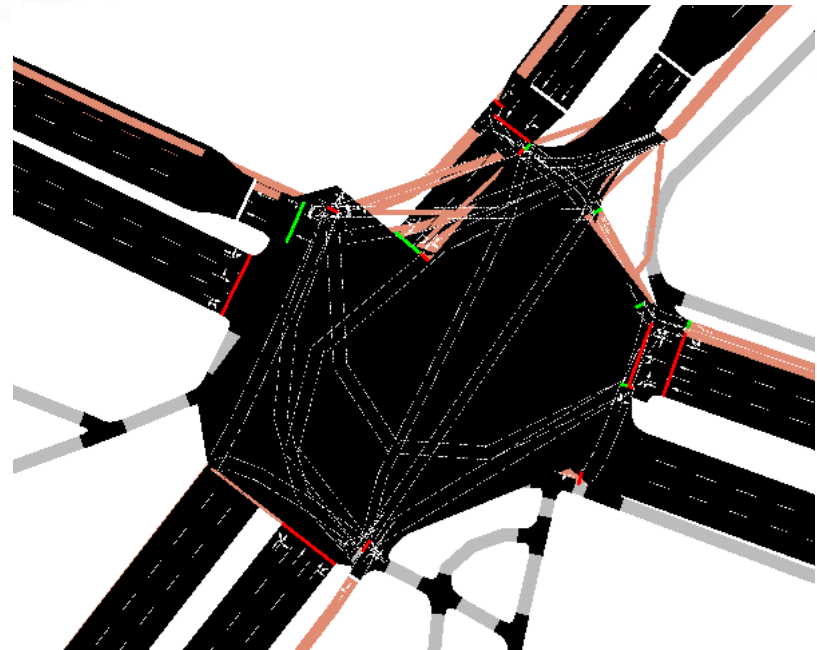
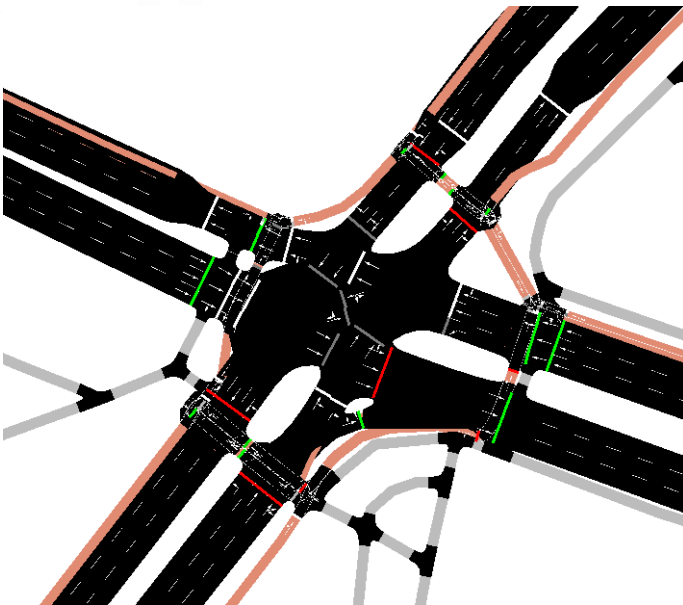
Topology

- Relevant roads and junctions must be present
 - Filtering road types
 - Filtering traffic modes
 - Roads under construction
- Junctions must be "joined" correctly



Joining Example

- Automatic joins may fail with too-much and too-little
- Recheck netconvert warnings containing "join"



Attributes

- **Lane count:** in most cases input data provides an undercount (i.e. missing turn-lanes)
 - Requires manual validation
 - Best way to find problems is when investigating congestion
- **Speed:** often provided in source data, fairly easy to verify visually (color edges by speed)
 - Use more appropriate typemap (osmNetconvertUrbanDe.typ.xml)
- Option **--osm.annotate-defaults** can help to find spots for manual checking of lane counts and speeds

Connectivity

- Lane-Connection information often missing in data source (i.e. OSM)
- Known issues when guessing connections (i.e. at slip-lanes)
- Option **--osm.turn-lanes** helps but doesn't cover all intermodal cases
- Common way to find problems is when investigating congestion
- Tooling
 - Netedit connection mode: select dead-ends / dead-starts
 - Sumo-gui: select reachable, color-by-reachability
 - **netcheck.py --right-of-way**
 - **netcheck.py --component-output**
 - **netconvert --keep-edges.components**

Access

- Access is often modelled with good quality in OSM
- For complex situations, option **--osm.lane-access** helps but doesn't cover all intermodal cases
- Problems can usually be found when investigating congestion

Right-of-way

- When no traffic lights are defined for an intersection it defaults to "priority"
 - By default, right of way is given to the road with the highest road class (i.e. highway.primary before highway.secondary)
 - Can be customized by modifying edge attribute 'priority'
- "small" junctions are set to right-before-left (configurable with option **--junctions.right-before-left.speed-threshold**)
- OSM data on right-of-way is available and could be used instead of default priorities (**not yet implemented**)

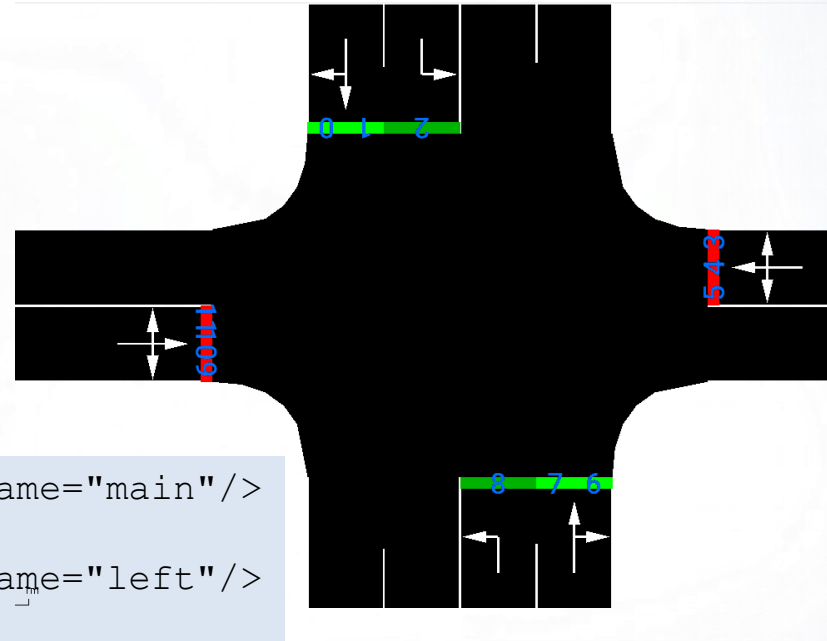
Infrastructure

- **Public transport stops**
 - can be automatically imported from OSM with option **--ptstop-output**
 - Recommended to use with option **--ptline-output** because this validates stop positions with known public transport routes
- **Fixed traffic counters**
 - Not generally available in OSM
 - Best to import from other sources: [mapDetectors.py](#)

Traffic Lights

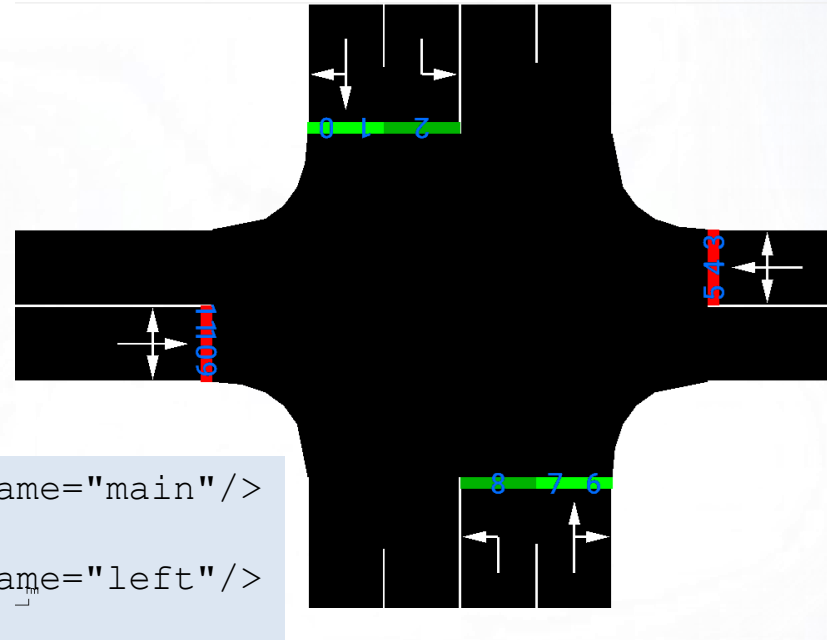
- No signal plans or timings available in public data sources
- Must be heuristically found for complex intersection
 - Option **--osm.guess-signals**
 - Option **--tls.join**
 - Option **--tls.guess**
- **Recommended manual check for presence of traffic lights at important intersections**

SUMO Idiosyncrasies



- Traffic light state is made explicit for each lane-to-lane connection (by default)
- Green light with/without conflict is made explicit ('g' , 'G')
- No distinction between phases and phase transitions
 - Each unique combination of signal states is encoded as a <phase>
 - Phases can be named for convenience
- Default cycle plans ignore all-red phase (changed by option --tls.allred.time)
- Cycle definitions are grouped by common state rather than by signal (group)

SUMO Idiosyncrasies



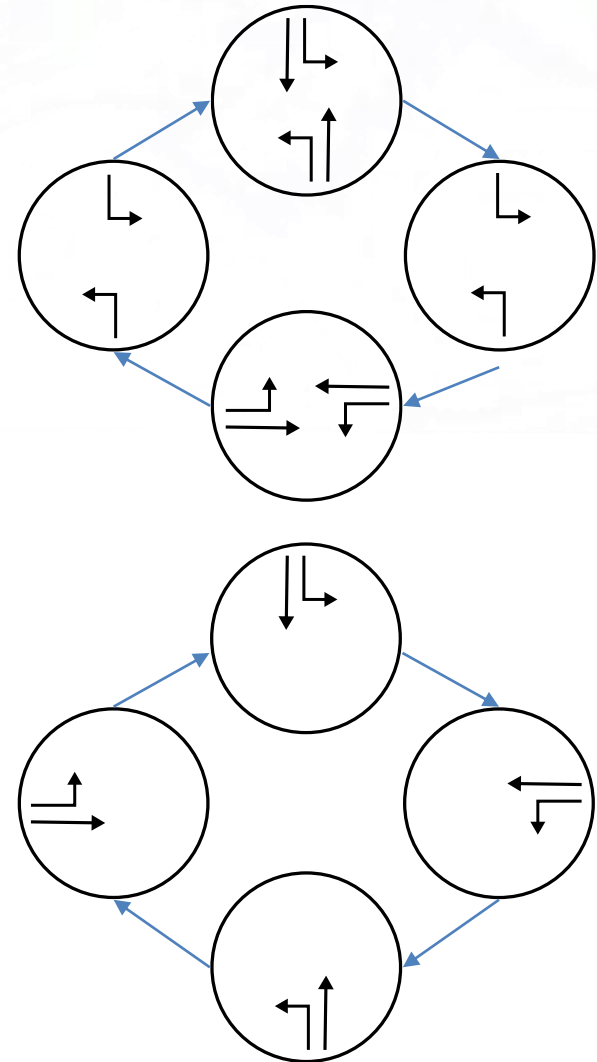
```
<phase duration="38" state="GGgrrrrGGgrrrr" name="main"/>
<phase duration="3" state="yygrrrryygrrrr"/>
<phase duration="6" state="rrGrrrrrrGrrr" name="left"/>
<phase duration="3" state="rryrrrrrryrrr"/>
<phase duration="37" state="rrrGGgrrrrGGg" name="side"/>
<phase duration="3" state="rrryyyyrrryyy"/>
```



Phase Layouts

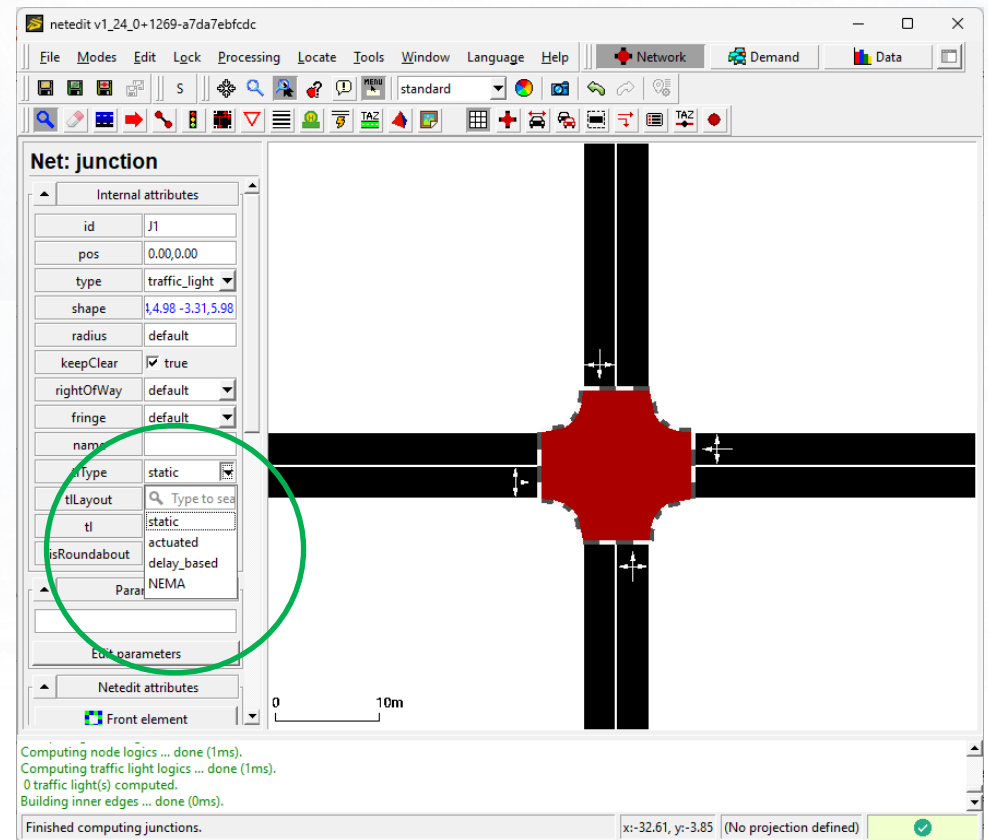
configured by junction attribute 'tLLayout'

- **'oncoming' (default)**: one direction of traffic has green together with the oncoming traffic (followed by a switch to side directions)
 - Left-turn phases are added automatically in the presence of left-turn-lanes
- **'incoming'**: each incoming road gets an exclusive green phase for all connections, roads are served in clockwise order



Actuated Traffic Lights

- Aborts phases after configurable detection delay on all served lanes
 - Detectors may be customized
- Runs through phases in fixed order (default programs)
- Supports branching programs

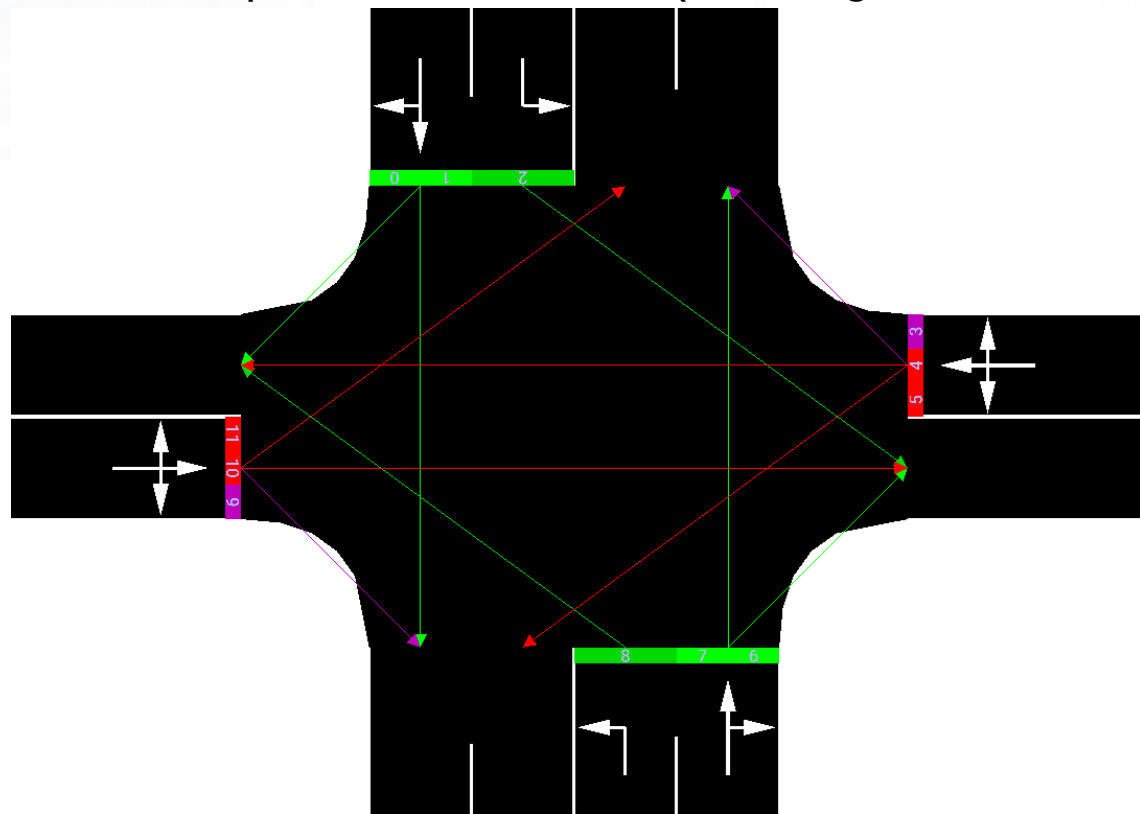


```
<tlLogic id="C" type="actuated" programID="0" offset="0">
  <phase duration="38" state="GGrrrrGGrrrr" minDur="5" maxDur="50"/>
  <phase duration="3" state="yygrrryygrrr"/>
  <phase duration="6" state="rrGrrrrrrGrrr" minDur="5" maxDur="50"/>
  <phase duration="3" state="rryrrrrrryrrr"/>
  <phase duration="37" state="rrrGGrrrrGGg" minDur="5" maxDur="50"/>
  <phase duration="3" state="rrryyyrrryyy"/>
</tlLogic>
```

Junction type traffic_light_right_on_red

- Modified right-of-way matrix to handle right-on-red driving
- Supports static and actuated programs
- Right turns have their red-phase state replaced with state 's' (denoting a stop-sign)

	dur	state
0	38.00	GGgsrrGGgsrr
1	3.00	yygsrryygsrr
2	6.00	rrGsrrrrrGsrr
3	3.00	rrysrrrrrrysrr
4	37.00	srrGGgsrrGGg
5	3.00	srryyysrryyy
Σ	90.00	Links: 12



SUMO-GUI

View Settings

custom_1

Background Color: uniform Interpolate

Streets

Vehicles waterway

Persons railway

Containers

Junctions Recalibrate Rainbow

Additional ☐ Draw with constant size when zoomed out

POIs Minimum Size 1

Polygons ☒ Draw junction shape

Legend ☒ Show lane to lane connections

OpenGL ☒ Show link tls index Size 120

☐ Show link junction index Size 50

☐ Show junction name Size 60

☐ Show internal junction name Size 50

☐ Show internal edge name Size 45

☐ Show crossing and walkingarea name Size 60

☒ Show traffic light phase Size 150

OK Cancel

tlLogic:C

Center

Switch off

Track Phases

Show Phases

Copy name to clipboard

Copy typed name to clipboard

Add To Selected

phase: 2

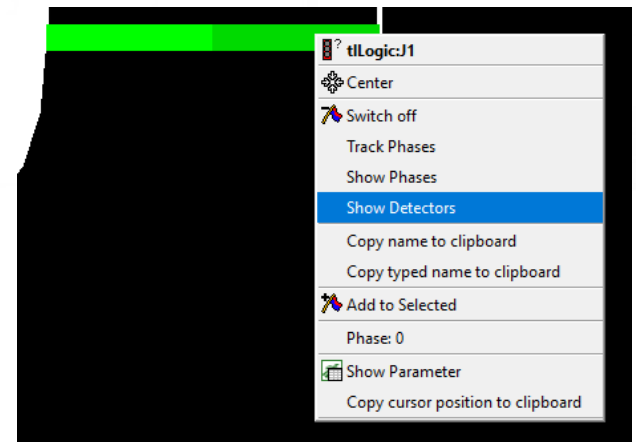
Show Parameter

Copy cursor position to clipboard

SUMO-GUI

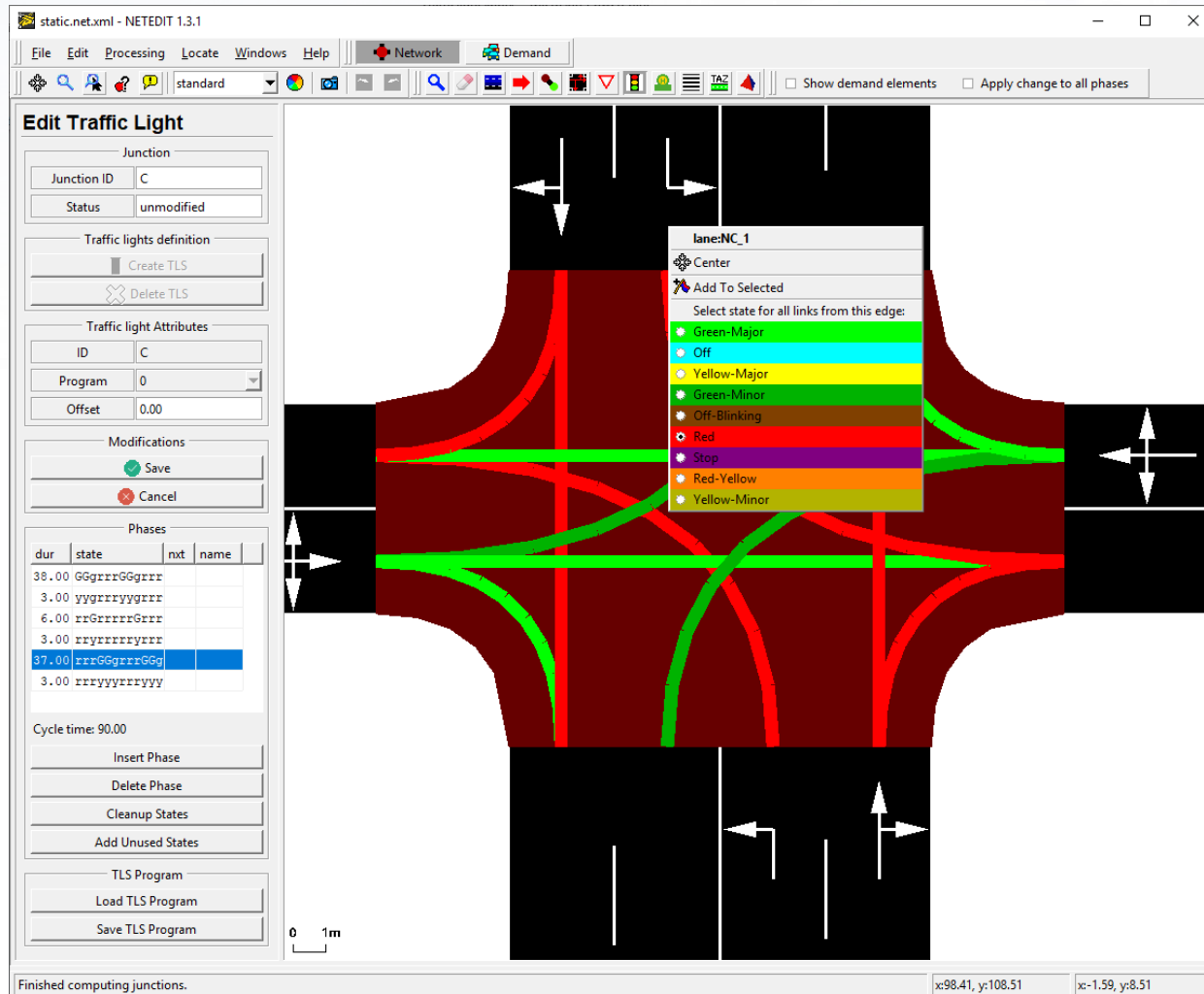
- Detectors for actuated traffic lights are generated automatically and hidden by default
 - Can be activated in sumo-gui (right-click on tls / colored bar)
- Can be activated by sumo option
 - **--tls.actuated.show-detectors**
- Can be activated by TLS param

```
<tlLogic id="C" type="actuated" ...">  
  <param key="show-detectors" value="true"/>  
  ...  
</tlLogic>
```



- Detector outline colors indicate:
 - Red: phase has been requested
 - Orange: jam detected
 - Green: detector is active for prolonging the current phase
 - White: detector is not used in this phase

Traffic lights - Netedit



Branching Phase Diagrams

- Phases may have multiple candidate successors
 - Implies: Phases can be skipped
- Supported by traffic light type 'actuated'
 - Modelled with phase attribute 'next'
 - Can also be modelled with external control (TraCI / libsumo)
- Two modes of operations for decision logic (can be mixed)
 - Automatic (only list candidate targets)
 - Custom (customize conditions for switching into candidate successor phase)